

iZiiApp Mushroom Production Architecture

Optimized for Costa Mushroom M2 Facility · 33 Grow Rooms

Target Audience: Plant Leadership & Operations Directors

Platform Version: 1.0.0
(Offline-First Edge)

Scope: Cultivation, Safety, Harvest & Labor

1. Executive Summary & Value Proposition

iZiiApp delivers an enterprise-grade, edge-first mobile and kiosk software platform built specifically for high-humidity, RF-shielded agricultural facilities like the **Costa Mushroom M2 plant**. Operating seamlessly with or without factory Wi-Fi, iZiiApp coordinates 33 grow rooms, automated worker safety protocols, and daily yield harvesting.

100% Offline Edge

Zero operational downtime. Local SQLite + P2P BLE mesh enables full function inside reinforced grow sheds without network latency.

Solo Worker Safety

Life-critical alone worker protection: automated countdown timers, CO/CO₂ thresholds, in-app alarms, and supervisor escalations.

Precision Harvest

End-to-end yield surveys, color team allocations, automated box target distribution, and standardized cycle tracking.

2. Edge-First Enterprise Topology

The system architecture separates physical plant floor operations from central IT dependencies using local edge nodes and auto-syncing mobile units.

COSTA M2 DISTRIBUTED EDGE ARCHITECTURE

1. Floor Edge Devices

Wall Kiosks & Terminals
NFC Scanner & Shift Display

Supervisor Mobiles
Flutter App + Local SQLite

Badges & Gas Probes
Staff Badges, CO / CO₂ Probes

2. Local iZii Engine & Mesh

P2P BLE Sync Mesh
Noise Protocol Cryptography

Drift SQLite Local DB
Full ACID Local Persistence

Outbox Sync Engine
Guaranteed Delivery Queue

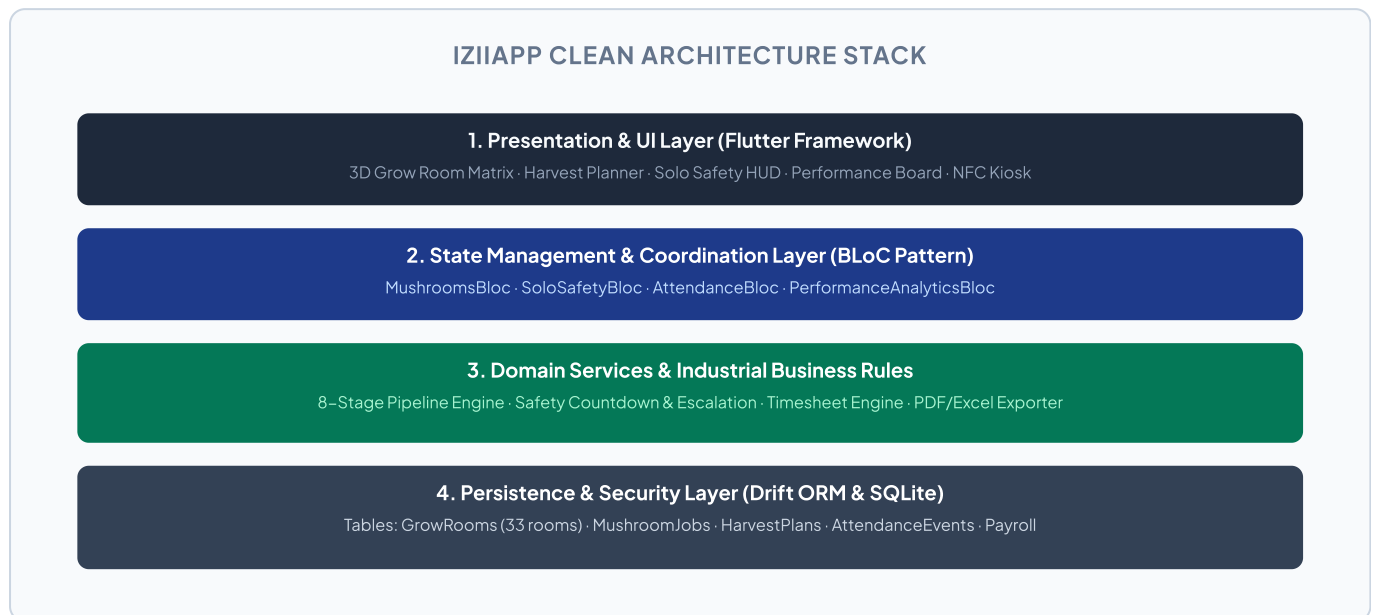
3. Central Plant Hub & ERP

Farm Central Server
Consolidated Backups & History

ERP & Payroll Connector
Automated Excel/PDF Exports

3. Layered Software Architecture (Mushroom Subsystem)

The Mushroom module adheres to high-cohesion, loose-coupling design principles, ensuring sub-second response times and rock-solid reliability across all UI screens.



4. Core Operational Pillars in Costa M2 Plant

4.1. 33-Room Cultivation Lifecycle (The 8-Stage Standard Pipeline)

Every grow room follows a strictly timed, quality-controlled pipeline. Custom recipe settings (e.g. Watering 2 Side 2L/m², Prochloraz 1.3g/m²) are validated at each stage transition:



4.2. Solo Worker & Alone-Worker Safety Protocol

Confined grow rooms with elevated CO₂ or chemical treatments require absolute safety adherence. iZiiApp enforces an automated lifecycle that alerts the worker and escalates to supervisors if a check-in is missed.

SOLO SAFETY ESCALATION LOGIC



4.3. Harvest Planning, Team Colors & Precision Logistics

iZiiApp connects pre-harvest **Yield Surveys** (Button, Cup, Flat strains) with daily retailer orders to generate actionable shift plans:

- **Color-Coded Picker Squads:** Ivory, Pearl, Purple, Sapphire squads allocated according to required pick-speed curves.
- **Trolley & Quality Directives:** Specific instructions (SSA – Single Stemming, Clumps removal) dynamically linked to each grow room.
- **Auto-Generated Shift Documents:** Single-click export of the Daily Job Plan PDF and Excel spreadsheets for factory floor notice boards.

4.4. Smart Attendance, NFC Badging & Break Policy Control

Workers tap their NFC badges on plant kiosks during entry, breaks, and departure. The timesheet engine automates break deduction policies (e.g. 30 min standard + 5 min grace), flag abnormal overtime, and prepares pre-payroll calculations for management review.

5. Executive Performance Metrics & Operational Benchmarks

Operational KPI	Standard Target	Business Impact for Management
On-Time Job Completion Rate	≥ 92.0%	Guarantees climate and watering timing, eliminating mushroom quality degradation.
Actual vs. Plan Duration Variance	± 5.0%	Detects equipment failure or labor under-allocation before daily bottleneck occurs.
Break Compliance Rate	≤ 35 mins	Enforces transparent labor compliance and eliminates unapproved overtime costs.
Solo Safety Unhandled Incidents	0 Incidents	100% auditable workplace safety record for WorkCover & SafeWork regulatory bodies.
Yield Forecast Accuracy	< 3.0% Error	Maximizes supermarket delivery fulfillment and prevents over/under picking penalties.

6. Role-Based Governance & Security Matrix

Role	Core Permissions & Operational Responsibilities
Plant Director / General Manager	Plant-wide executive dashboards, cross-department KPI analytics, payroll sign-off, system policy setup.
Shift & Operations Supervisor	Daily harvest plan publishing, job reassignments, on-time overrides, solo safety emergency resolution.
Growing Specialist	8-step cycle execution, recipe parameter entry (prochloraz, watering volume), CO/CO ₂ gas logging.
Harvest Team Leader	Picker team supervision, room trolley dispatch, real-time box counting and quality verification.
Picker / Floor Operator	NFC check-in/out, break logging, interactive task checklist confirmation.

7. Strategic ROI & Leadership Conclusion

Key ROI Deliverables:

- **15–20% Reduction in Supervisory Overhead:** Automated shift assignment and PDF/Excel generation save up to 2 hours per shift.
- **100% Lone Worker Compliance:** Full audit trail of gas readings and check-in logs protects the business from safety liability.
- **Maximum Crop Yield Consistency:** Strict 8-step adherence delivers higher crop weight and uniform button mushroom grading.